

Pick and Place SOP

Setup

Complete both checklists below before starting any episode.

Hardware checklist

- Cameras are on and recording
- Env camera frame includes the front and back edges of the table and is centered on the table's midpoint
- Gripper is at the home position with gripper open
- Table surface is clear except for the cup/object and the tray/target

Materials checklist

- Source area contains N objects (define type/size) arranged within reachable workspace
- Target area (bin/slot/marked region) is empty and reachable
- Any fixtures (tray, jig, dividers) are secured and aligned to the table area.
- If using labels/markers: markers are visible and not occluded (in line of sight or not blocked)

Workspace layout

- Left / Source → objects to pick (input)
 - Center → transit/working area (optional)
 - Right / Target → placement area (output)
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Steps

Step 1: Move to object and grasp (pick)

Goal: N object is securely grasped and lifted clear of the table.

- Move the gripper from home position to the object.
- Approach from the top or the side, using a smooth, collision-free path.
- Grasp securely around the object body (or handle if present).
- Lift the object clear of the table and pause briefly to confirm it is stable (no slip/rotation).

Step 2: Adjust / re-grasp (if needed)

Goal: secure, centered grasp suitable for precise placement.

- If the grasp is unstable, off-center, or slipping:
 - Lower the object gently onto the table in the center/working area.
 - Release and re-grasp with better finger placement.
 - Re-lift and confirm the object is held firmly.
- If the grasp is stable, skip this step and continue.

Step 3: Transport and place on the target

Goal: object is placed upright/centered and stable at the target location.

- Carry the object from the source area to the target (e.g., tray/bin/marked region).
- Hover above the target and visually align position and orientation.
- Lower slowly and precisely until the object is fully seated/on the surface.
- Release only when the object is stable and not moving or shaking unsteadily.
- Lift the gripper slightly away from the object.

Step 4: Return to home and end the episode

- Return the gripper to the home position with the gripper open.
- End data collection/recording for the episode.

After each episode: repeat or reset

This section is not a recorded subtask: it covers session-level workflow between episodes.

- If more objects remain in the source area → continue to Step 1 with the next object.
 - If the source area is empty or the target area is full → reset the workspace:
 - With recording off, clear the target area as needed.
 - Refill/rearrange the source area to the initial setup state.
 - Re-verify setup checklists before the next run.
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Annotation subtasks

1. Move to object and grasp (pick) (Step 1)
 2. Grasp adjustment / verification (Step 2)
 3. Transport and precise placement (Step 3)
 4. Return to home (Step 4)
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SOP Rubric (Steps-only violations)

Use this rubric to score whether each episode followed the SOP steps. These are process/step adherence checks (not quality-of-fold outcome checks), and each item is a violation if it occurs at any point during the episode.

01: Move to object smoothly

Gripper moves from home to the object using a smooth, collision-free path (top/side approach).

02: Secure grasp

Object is grasped securely around the body/handle (not pinched precariously or off-contact).

03: Lift clear + stable hold

Object is lifted clear of the table and remains stable during a brief pause (no slip/rotation).

04: Correct decision on re-grasp

If grasp is unstable/off-center/slipping, re-grasp is attempted; if grasp is stable, step is correctly skipped.

05: Re-grasp executed correctly (when needed)

Object is lowered to center/working area, released, re-grasped with improved finger placement, then re-lifted stable.

06: Controlled transport + alignment

Object is carried to target without collisions/drops; robot hovers above target and aligns position/orientation before lowering.

07: Precise placement + clean release

Object is lowered slowly until seated; release occurs only when stable; gripper lifts away without disturbing the object.

08: Return to home

Gripper returns to the home position with gripper open.